[TODO: title]

[TODO: authors and IEEE membership]

***Abstract*— [TODO: abstract (write last): one paragraph, 150–250 words, no abbreviations, references, equations or tables (JSEN template rule)]**

***Index Terms*— Foundation models, missing modality, multimodal semantic segmentation, robustness, sensor fusion**

todo: graphical abstract (required by the JSEN template, 3 in wide)

TABLE I
DELIVER (RGB, DEPTH, EVENT, LIDAR), MIOU (%).

Method	Backbone	Sensors	val	test
CMNeXt todo	todo	RGB-D-E-L	todo	todo
StitchFusion todo	todo	RGB-D-E-L	todo	todo
GeminiFusion todo	todo	RGB-D-E-L	todo	todo
CAFuser todo	todo	RGB-D-E-L	todo	todo
CAFuser-CAA todo	todo	RGB-D-E-L	68.79	todo
DGFusion todo	Swin-T	RGB-D-E-L	todo	56.71
Fewer sensors (secondary comparison)				
MM SAM-adapter todo	todo	RGB-D	69.60	57.35
Ours (3-seed mean)	DINOv3-L	RGB-D-E-L	todo	57.74
Ours (best seed)	DINOv3-L	RGB-D-E-L	todo	58.61

todo: original table/page per baseline; protocol (val-best checkpoint, 1024, nearest-exact label resampling); DINOv3-L is frozen.

TABLE II
MUSES, OFFICIAL EVALUATION, MIOU (%).

Method	Backbone	Sensors	val	test
CAFuser todo	todo	todo	todo	todo
DGFusion todo	todo	todo	todo	79.5
Camera only or fewer sensors (secondary comparison)				
GtA todo	todo	RGB	todo	82.39
MM SAM-adapter todo	todo	RGB-L	todo	81.07
Ours	DINOv3-L	RGB-E-L-R	todo	todo

TABLE III
MCUBES (RGB, AOLP, DOLP, NIR), TEST MIOU (%).

Method	Backbone	test
Mul-VMamba todo	todo	54.65
StitchFusion todo	todo	55.9
Ours	DINOv3-L	todo

I. Introduction

[TODO: 0.8 page: problem, two tracks (accuracy vs. missing-sensor robustness), three contributions]

II. Related Work

[TODO: 0.6 page: all-modality fusion; missing-modality robustness; frozen foundation-model adaptation]

III. Sensors, Benchmarks, and Protocol

[TODO: 0.5 page: sensor physics and failure modes; DELIVER/MUSES/MCubeS; evaluation protocol]

IV. Method

[TODO: 1.5 pages: architecture figure, two-pass training with teacher distillation, high-resolution path]

V. Experiments

[TODO: 2.5 pages: implementation details (frozen DINOv3-L, LoRA r8, 40-epoch schedule, val-best checkpoint, legal v2 at 1024), then Tables 1–7]

[TODO: submission date; funding]
[TODO: affiliations and e-mail]

VI. Analysis

[TODO: 1.5 pages: where the sensor over-reliance comes from, source of the resolution gain, thin objects, failure case, limitations]

VII. Conclusion

[TODO: 0.3 page]

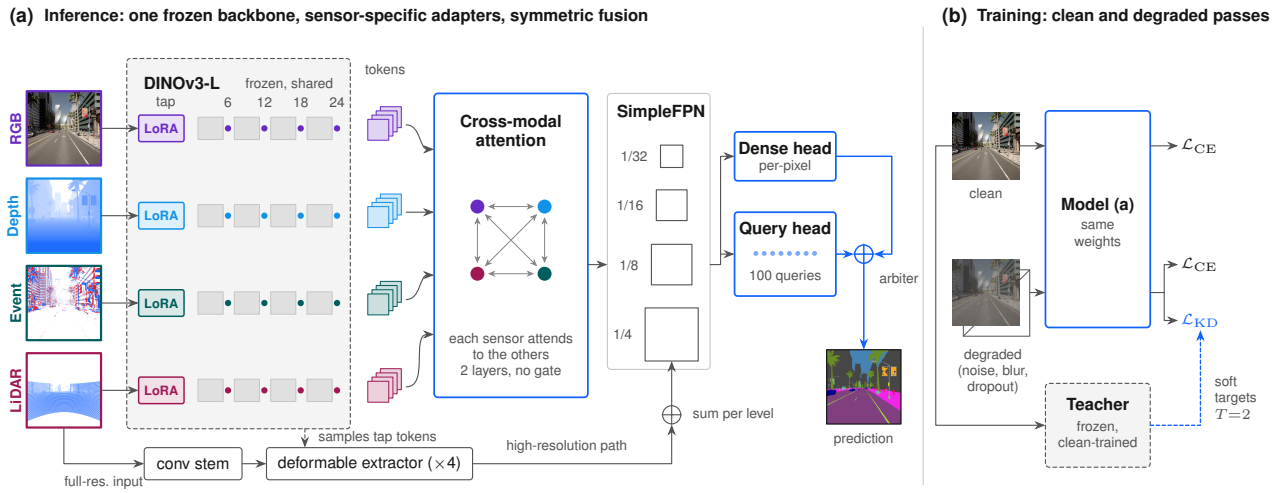


Fig. 1. Overview. (a) Each sensor passes through the same frozen DINOv3-L with its own LoRA adapter; tokens from four depths are fused by symmetric cross-modal attention, decoded by a SimpleFPN augmented with a high-resolution path, and classified by a dense head and a query head combined by an arbiter. (b) Every batch is used twice: a clean pass and a synthetically degraded pass, the latter also distilled from a frozen teacher trained on clean data. Blue outlines mark trained components; dashed boxes are frozen. [TODO: draft drawn by the paper session; redraw per notes/figure_guide.md]

TABLE IV
ROBUSTNESS TO MISSING AND DEGRADED SENSORS ON DELIVER, MIOU (%).

(a) Public missing-modality benchmark, val

Method	EMM	RMM .25	RMM .50	RMM .75
RMMSS todo	49.93	todo	todo	todo
Ours (3 seeds)	57.85	65.53	63.35	61.28

(b) Same protocol for all models, test

Method	Clean	EMM	S&P .05	S&P .10	S&P .20
DGFusion (official)	55.68	31.53	37.97	16.71	5.52
DGFusion (retrained†)	54.77	45.52	48.89	40.51	23.34
CAFuser (retrained†)	55.79	46.50	todo	todo	todo
Ours w/o two-pass	todo	40.01	todo	todo	todo
Ours w/o high-res	56.59	48.06	53.25	51.64	48.79
Ours	57.74	48.66	53.11	51.36	47.85

†Retrained with the same synthetic degradation. Ours: mean of 3 seeds; baselines: single runs. todo: define EMM (mean over 15 sensor subsets), RMM (random drop ratio), S&P noise.

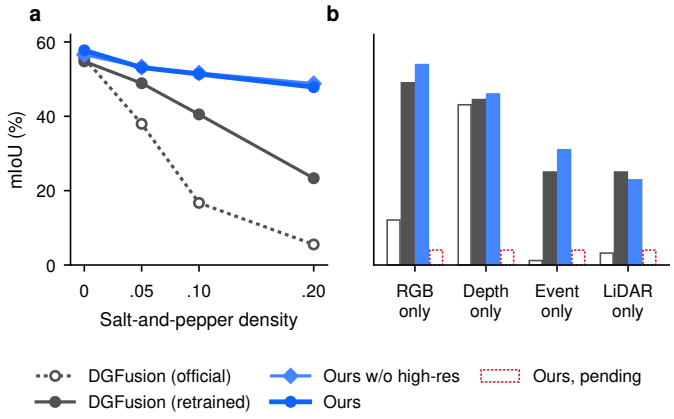


Fig. 2. Robustness on DELIVER test under one protocol. (a) Salt-and-pepper noise applied to every sensor. (b) Only one sensor available (the other three zero-filled). Ours: mean of three seeds; DGFusion: single runs, official weights or retrained with the same synthetic degradation. [TODO: (b) full-model values pending.]

TABLE V
DELIVER TEST MIOU (%) PER CONDITION, ALL SENSORS PRESENT.

Model	Sun	Cloud	Rain	Fog	Night	Low exp.	All
DGFusion (official)	todo	todo	todo	todo	todo	todo	55.68
DGFusion (retrained†)	todo	todo	todo	todo	todo	todo	54.77
CAFuser (retrained†)	todo	todo	todo	todo	todo	todo	55.79
Ours w/o high-res	todo	todo	todo	todo	todo	todo	56.59
Ours	todo	todo	todo	todo	todo	todo	57.74

†Retrained with synthetic degradation. All: same values as Table IV(b) clean column.

TABLE VI
ABLATION ON DELIVER TEST (ONE SEED, 40 EPOCHS), MIOU (%).

Configuration	Clean	EMM
Building up		
Frozen DINOv3-L + LoRA, RGB only	55.14	–
+ depth, event, LiDAR (mean fusion)	todo	todo
+ cross-modal attention, query head	55.94	todo
+ two-pass training, distillation	57.73	48.68
+ high-res path (full model)	58.61	49.12
Removing one component from the full model		
− sensor-specific LoRA (one shared)	58.84	todo
− cross-modal attention (mean fusion)*	57.76	todo
− query head and arbiter	57.89	todo
− degradation pass	todo	todo
− teacher distillation	todo	todo

*Also removes the trunk expansion. todo: EMM column for ablation runs; decision on shared LoRA pending its robustness test.

TABLE VII
COMPUTATIONAL COST ON DELIVER AT 1024×1024.

Model	Params (M)	Trainable (M)	GFLOPs	Latency (ms)	Memory (GB)
DGFusion	todo	todo	todo	todo	todo
CAFuser	todo	todo	todo	todo	todo
Ours w/o high-res	todo	todo	todo	todo	todo
Ours	todo	todo	todo	todo	todo

todo: Fig. 3 when a sensor fails: rows = depth missing / RGB missing / LiDAR+event missing / RGB with heavy noise; columns = affected input, ground truth, DGFusion (official), DGFusion (retrained), CAFuser (retrained), ours; error pixels in one colour

Fig. 3. [TODO: caption; state the scene-selection rule (top differences between ours and the best baseline, one scene per condition, no repeated map)]

todo: Fig. 4 adverse conditions, all sensors present: DELIVER night / fog / rain / low exposure and MUSES real night / snow / fog; columns = RGB, ground truth, DGFusion, CAFuser, ours; zoomed crops where the baselines miss objects

Fig. 4. [TODO: caption; state the scene-selection rule]

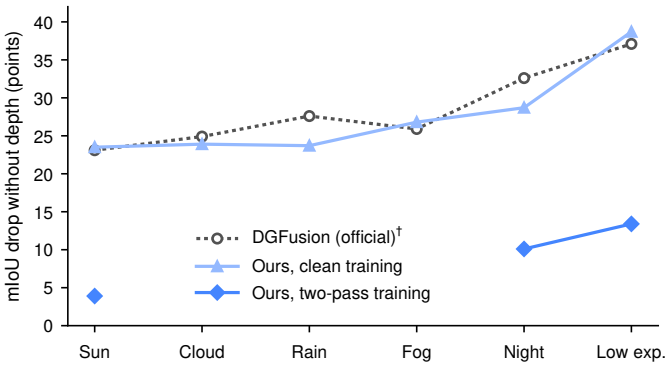


Fig. 5. Drop in mIoU when depth is removed, per condition. A model trained only on clean data relies on depth as heavily as DGFusion, whose design adds depth supervision; two-pass training reduces the drop to a third or less with the same architecture. †Measured on test; the other curves on val. [TODO: re-measure all curves on one split; add the full model and the retrained baselines.]

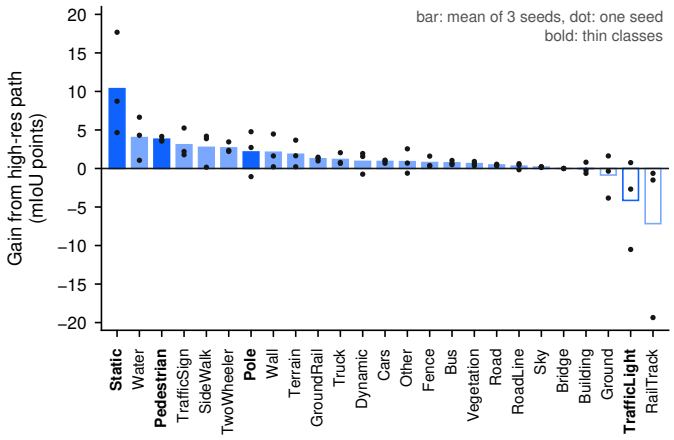


Fig. 6. Per-class gain of the high-resolution path on DELIVER test, paired by seed (full model minus the same model without the path). Thin classes gain most; TrafficLight and RailTrack lose and vary strongly across seeds (Section VI). [TODO: add zoomed crops of thin objects (Q2 dumps on yeon).]

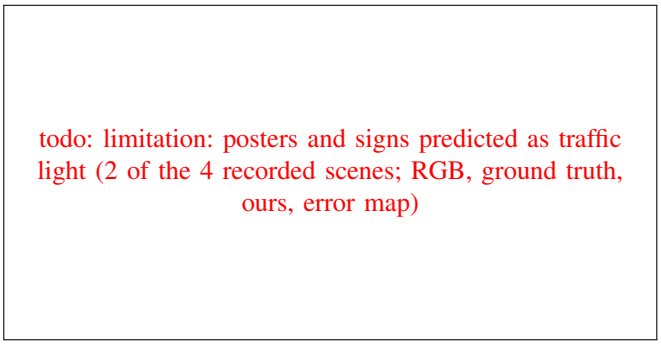


Fig. 7. [TODO: caption; selection rule from the scene README]